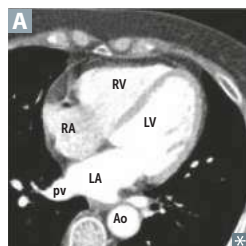


► CARDIOVASCULAR—ANATOMY

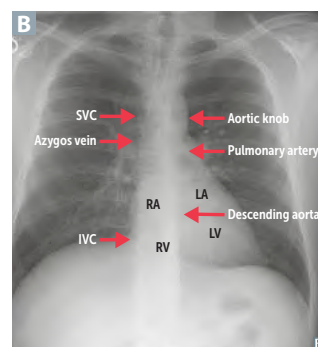
Heart anatomy



LA is the most posterior part of the heart **A**; LA enlargement (eg, in mitral stenosis) can lead to:

- Compression of esophagus → dysphagia
- Compression of left recurrent laryngeal nerve (branch of vagus nerve → hoarseness) (**Ortner syndrome**)

RV is the most anterior part of the heart and most commonly injured in trauma. LV is about 2/3 and RV is about 1/3 of the inferior (diaphragmatic) cardiac surface **B**.



Pericardium

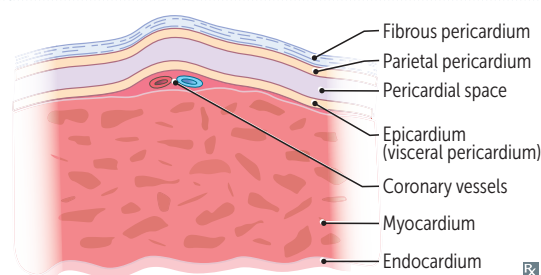
Consists of 3 layers (from outer to inner):

- Fibrous pericardium
- Parietal pericardium
- Epicardium (visceral pericardium)

Pericardial space lies between parietal pericardium and epicardium.

Pericardium innervated by phrenic nerve.

Pericarditis can cause referred pain to the neck, arms, or one or both shoulders (often left).



Coronary blood supply

LAD and its branches supply anterior 2/3 of interventricular septum, anterolateral papillary muscle, and anterior surface of LV. Most commonly occluded.

Posterior descending artery (PDA) supplies posterior 1/3 of interventricular septum, posterior 2/3 of ventricular walls, posteromedial papillary muscle, and SA and AV nodes (as determined by dominance). Infarct may cause nodal dysfunction (bradycardia or heart block).

Right (acute) marginal artery supplies RV.

Dominance:

- Right-dominant circulation (most common) = PDA arises from RCA
- Left-dominant circulation = PDA arises from LCX (~5%–10% of patients)
- Codominant circulation = PDA arises from both LCX and RCA (~10%–20% of patients)

Coronary blood flow to LV and interventricular septum peaks in early diastole.

Coronary sinus runs in the left AV groove and drains into the RA.

Key:
 AMA = Acute marginal artery
 LAD = Left anterior descending artery
 LCA (or LM) = Left (main) coronary artery
 LCX = Left circumflex artery
 OMA = Obtuse marginal artery
 PDA = Posterior descending artery
 PT = Pulmonary trunk
 PV = Pulmonary vein
 RCA = Right coronary artery

